PROJECT DOCUMENT

BIOMASS ENERGY & AI

Reforesting in Mitsue

Earned Value Management Plan

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BIOMASS ENERGY & AI — Earned Value Management Plan

Reforesting in Mitsue

Performance Baseline · Cost Control · Forecast Framework

Phases 0–3 · April 2026 – September 2028

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1. Purpose and Scope

This EVM Plan establishes the **Performance Measurement Baseline (PMB)** for the Mitsue Sustainable Energy & AI Data Center project across its four build phases (Phase 0 through Phase 3). It defines how the project will measure schedule and cost performance, report progress to stakeholders, and forecast the cost and date at completion.

The plan covers the **30-month period April 2026 – September 2028**. Phase 4 (Operations & Scale, Month 31+) is not included in the baseline because it is funded differently (operating revenue) and planned in Year 2 once Phase 1 results are confirmed. *The operating revenue model and capital-payback analysis that fund Phase 4 are set out in `mitsue_revenue_model.md` and the Implementation Plan's expanded Phase 4 section — see those documents; this PMB deliberately stops at the end of Phase 3 (construction). The month-by-month funding **inflow** that must keep the cash balance positive against this spend baseline is set out in `mitsue_cashflow_model.md`.*

Data Date for current baseline: 2026-05-18 (end of Month 2)

> **Schedule reconciliation (2026-07-03).** This PMB is anchored to the original compressed schedule (M1 = Apr 2026, 30 months to Sep 2028). The **live OpenProject Gantt** — now the operative schedule — runs materially later: Phase 0 to Oct 2026, Phase 1 to May 2027, Phase 2 to May 2028, Phase 3 to Nov 2029 (~44 months). Phase 0 is on-track against the Gantt; the earlier sense of slippage came from comparing to this compressed baseline. The monthly PV curve and CPI/SPI figures below remain **Baseline Rev 1** and have **not** been re-timed to the Gantt — re-timing the PMB is the explicit **Baseline Rev 2** deliverable (due M9 / Dec 2026), which will also fold in any confirmed 交付金 amount. Until then, read schedule variance against the live Gantt, not this Rev 1 curve.

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2. EVM Fundamentals — Key Metrics

Term	Symbol	Formula	What it means
Planned Value	**PV**	Time-phased budget	Budgeted cost of work **scheduled** to date
Earned Value	**EV**	% complete × BAC	Budgeted cost of work **actually performed**
Actual Cost	**AC**	Invoices + payroll	Real money **spent** to date
Budget at Completion	**BAC**	Sum of all PV	Total approved budget for the PMB
Schedule Variance	**SV**	EV − PV	Negative = behind plan

Term	Symbol	Formula	What it means
Cost Variance	**CV**	EV − AC	Negative = over budget
Schedule Performance Index	**SPI**	EV ÷ PV	< 1.0 = behind schedule
Cost Performance Index	**CPI**	EV ÷ AC	< 1.0 = over budget
Estimate at Completion	**EAC**	AC + (BAC−EV)÷CPI	Projected final cost
Estimate to Complete	**ETC**	EAC − AC	Remaining work cost forecast
Variance at Completion	**VAC**	BAC − EAC	Projected over/under at end
To-Complete Perf. Index	**TCPI**	(BAC−EV)÷(BAC−AC)	Efficiency needed to finish on budget

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3. Budget Structure

3.1 Summary Budget

Level	Item	Budget (¥M)	% of BAC
Total Project Budget		**¥245.0M**	—
— Management Reserve (~11%)	Not in PMB	**¥25.0M**	—
Performance Measurement Baseline (BAC)		**¥220.0M**	100%
1.0	Project Management & Governance	¥15.0M	6.8%
2.0	Phase 0 — Pre-Foundation	¥0.25M	0.1%
3.0	Phase 1 — Foundation	¥5.5M	2.5%
4.0	Phase 2 — Pilot Design	¥22.25M	10.1%
5.0	Phase 3 — Pilot Build	¥177.0M	80.5%

> **Management Reserve** (¥25M) is held by the Representative Director and is **not part of the PMB**. It may be drawn down only with explicit board approval to address unforeseen scope.

> **Baseline Rev 1 (May 2026)** — Reality-checked against real-world Japan benchmarks for solar PV (¥200–300K/kW), school seismic retrofit (¥200–500K/㎡), commercial EV fast-chargers (¥5–6M each), and forestry road works. Phase 3 line items, PM/governance, and contingency uplifted accordingly; BAC raised from ¥168M to ¥220M.

3.2 Work Breakdown Structure (WBS) — Budget Detail

WBS	Element	Budget (¥M)	Phase
1.0	**Project Management & Governance**		

WBS	Element	Budget (¥M)	Phase
1.1a	Representative Director (compensation — phased; see §13)	¥5.0M	All
1.1b	Project coordinator / admin	¥3.0M	All
1.2	Legal, accounting, 行政書士 fees	¥4.0M	All
1.3	Communications, website, translation	¥3.0M	All
	Sub-total 1.0	*¥15.0M*	
2.0	**Phase 0 — Pre-Foundation**		
2.1	Community stakeholder meetings & travel	¥0.10M	P0
2.2	Charter & document preparation	¥0.10M	P0
2.3	Founding team identification	¥0.05M	P0
	Sub-total 2.0	*¥0.25M*	
3.0	**Phase 1 — Foundation**		
3.1	一般社団法人 incorporation	¥0.5M	P1
3.2	Forestry feasibility study	¥1.5M	P1
3.3	Energy systems feasibility study	¥1.5M	P1
3.4	Building & site assessment	¥0.8M	P1
3.5	Connectivity assessment	¥0.4M	P1
3.6	Advisory board, bank, accounting setup	¥0.8M	P1
	Sub-total 3.0	*¥5.5M*	
4.0	**Phase 2 — Pilot Design**		
4.1	Structural & architectural design	¥2.0M	P2
4.2	Energy systems engineering	¥2.0M	P2
4.3	Data center & IT design	¥1.0M	P2
4.4	EV charging system design	¥1.0M	P2
4.5	Permitting & regulatory (METI, FIT, forestry)	¥3.5M	P2
4.6	Partnership & landowner agreements	¥1.5M	P2

WBS	Element	Budget (¥M)	Phase
4.7	Grant writing & funding applications	¥2.0M	P2
4.8	Staff hiring & onboarding (2–3 part-time)	¥6.0M	P2
4.9	Vendor pre-qualification	¥1.5M	P2
4.10	Phase 2 contingency	¥1.75M	P2
	Sub-total 4.0	*¥22.25M*	
5.0	**Phase 3 — Pilot Build**		
5.1	School building renovation (1 wing)	¥38.0M	P3
5.2	Solar PV installation (~100 kW)	¥22.0M	P3
5.3	Battery storage system	¥12.0M	P3
5.4	EV charging infrastructure (4 stations)	¥15.0M	P3
5.5	Data center fitout (10–20 servers)	¥20.0M	P3
5.6	Forestry operations (5–10 ha harvest + replant)	¥25.0M	P3
5.7	Fiber connectivity upgrade	¥10.0M	P3
5.8	Testing, commissioning, and startup	¥8.0M	P3
5.9	Phase 3 contingency (18%)	¥27.0M	P3
	Sub-total 5.0	*¥177.0M*	
	TOTAL BAC	**¥220.0M**	

4. Time-Phased Baseline (S-Curve)

Monthly planned expenditure and cumulative planned value over the 30-month baseline. Spending is front-loaded within phases on studies and design; construction spending peaks in Months 22–27.

Month	Calendar	Phase	Monthly PV (¥M)	Cumulative PV (¥M)	% Complete
M1	Apr 2026	P0	0.05	0.05	0.0%
M2	May 2026	P0	0.10	0.15	0.1%
M2 ← Status Date					
M3	Jun 2026	P0	0.10	0.25	0.1%
M4	Jul 2026	P1	0.20	0.45	0.2%
M5	Aug 2026	P1	0.50	0.95	0.4%

Month	Calendar	Phase	Monthly PV (¥M)	Cumulative PV (¥M)	% Complete
M6	Sep 2026	P1	0.80	1.75	0.8%
M7	Oct 2026	P1	1.20	2.95	1.3%
M8	Nov 2026	P1	1.50	4.45	2.0%
M9	Dec 2026	P1	1.30	5.75	2.6%
M10	Jan 2027	P2	1.50	7.25	3.3%
M11	Feb 2027	P2	2.20	9.45	4.3%
M12	Mar 2027	P2	2.80	12.25	5.6%
M13	Apr 2027	P2	3.00	15.25	6.9%
M14	May 2027	P2	3.00	18.25	8.3%
M15	Jun 2027	P2	3.00	21.25	9.7%
M16	Jul 2027	P2	2.50	23.75	10.8%
M17	Aug 2027	P2	2.50	26.25	11.9%
M18	Sep 2027	P2	2.00	28.25	12.8%
M19	Oct 2027	P3	4.00	32.25	14.7%
M20	Nov 2027	P3	7.00	39.25	17.8%
M21	Dec 2027	P3	11.00	50.25	22.8%
M22	Jan 2028	P3	16.50	66.75	30.3%
M23	Feb 2028	P3	20.50	87.25	39.7%
M24	Mar 2028	P3	20.50	107.75	49.0%
M25	Apr 2028	P3	25.00	132.75	60.3%
M26	May 2028	P3	25.00	157.75	71.7%
M27	Jun 2028	P3	20.50	178.25	81.0%
M28	Jul 2028	P3	20.50	198.75	90.3%
M29	Aug 2028	P3	14.00	212.75	96.7%
M30	Sep 2028	P3	7.25	220.00	100.0%
	BAC		**¥220.0M**		

S-Curve Shape

The expenditure profile follows a classic **slow-fast-slow** S-curve:

- **M1–M9** (Phases 0–1): Ramp-up — legal, feasibility studies, advisory engagement
- **M10–M18** (Phase 2): Acceleration — engineering design, permitting, grant applications
- **M19–M28** (Phase 3 core): Peak spend — construction, procurement, installation
- **M29–M30** (Phase 3 close): Tail-off — commissioning, punch-list, handover

Figure 1 — Baseline S-Curve (Cumulative Planned Value)

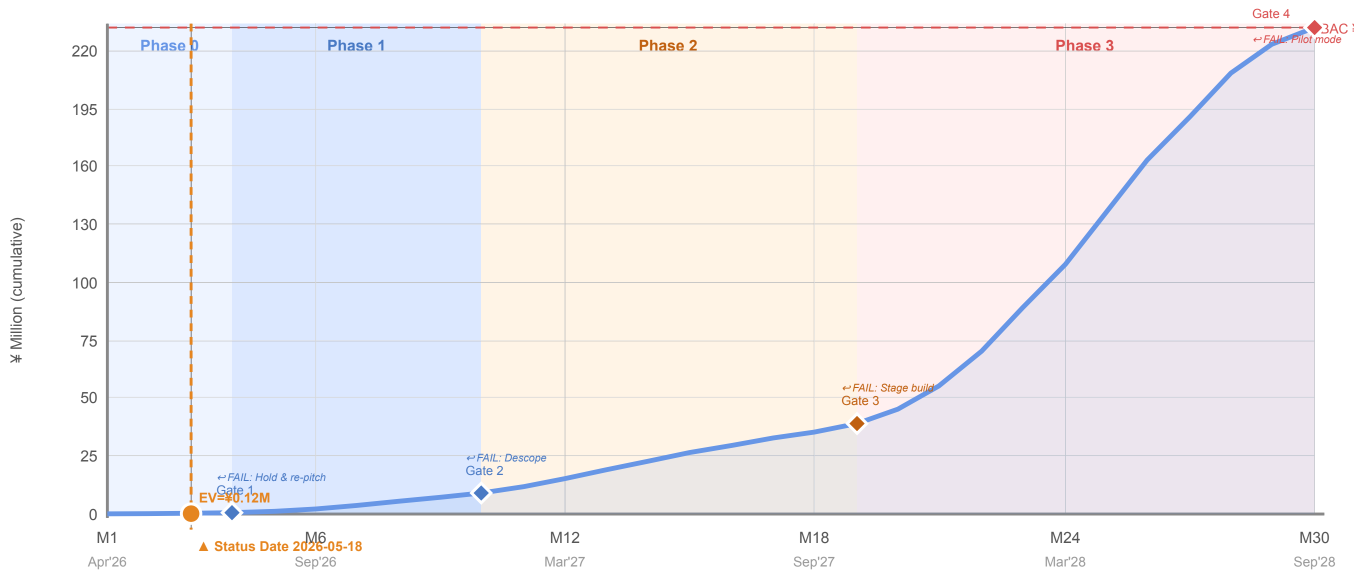
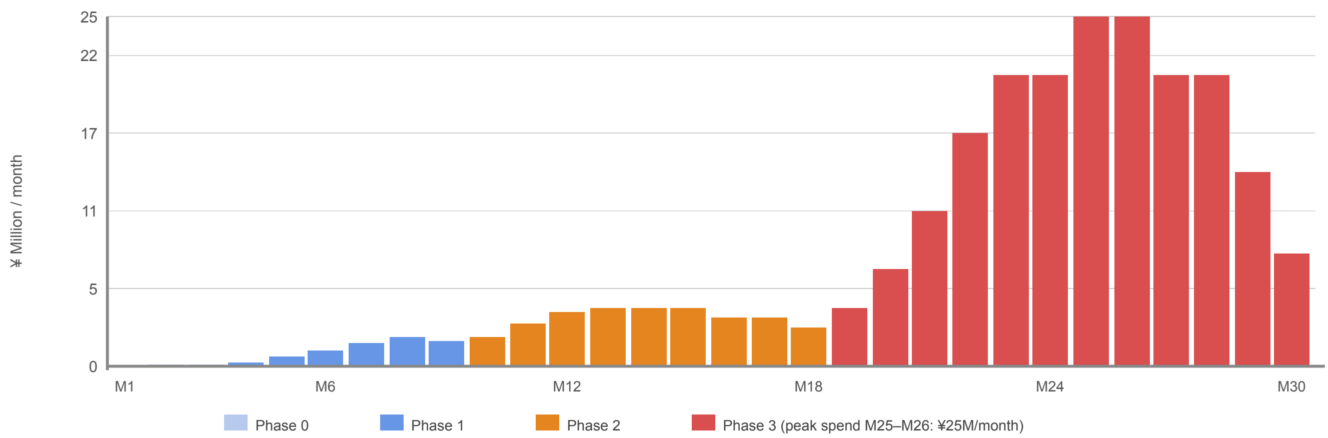


Figure 2 — Monthly Planned Expenditure by Phase



5. Current Performance Status

Data Date: 2026-05-18 (End of Month 2)

5.1 Status Narrative

The project is in **Phase 0 — Pre-Foundation**. Activities in progress include quiet stakeholder conversations with village leadership, preliminary charter drafting, and identifying founding team candidates. The Japanese co-founder search is underway but not yet resolved — this is the single most critical Phase 0 dependency.

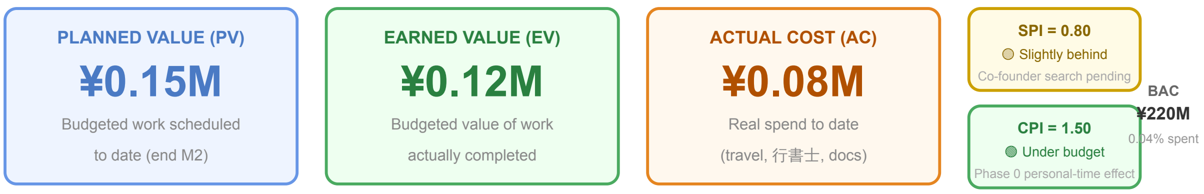
Expenditure to date is personal travel, document translation, and a preliminary consultation with a Nara-based 行政書士. No major contracts have been signed.

5.2 Performance Metrics — Phase 0

Metric	Value	Notes
PV (Planned Value to date)	¥0.15M	Planned spend through M2
EV (Earned Value to date)	¥0.12M	~80% of planned Phase 0 work done

Metric	Value	Notes
AC (Actual Cost to date)	¥0.08M	Under plan — mostly personal time
SV (Schedule Variance)	**−¥0.03M**	Slightly behind: co-founder not yet identified
CV (Cost Variance)	**+¥0.04M**	Under budget: Phase 0 costs lower than planned
SPI	**0.80**	80% of planned work achieved on schedule
CPI	**1.50**	Each ¥1 spent is delivering ¥1.50 of value

Figure 3 — Current EVM Performance Dashboard (Data Date: 2026-05-18)



> **Note:** CPI of 1.50 at Month 2 is partly a Phase 0 artefact — most value creation is through personal time rather than budgeted spend. Do not extrapolate CPI to total project forecast until Phase 1 feasibility contracts are in place.

5.3 Phase 0 Forecast

Metric	Value
Phase 0 BAC	¥0.25M
EAC (Phase 0) = AC + (BAC−EV)÷CPI	¥0.16M
ETC (Phase 0) = EAC − AC	¥0.08M
VAC (Phase 0) = BAC − EAC	**+¥0.09M under budget**
TCPI	0.87 — need only 0.87 efficiency to finish Phase 0 within budget

6. Phase-by-Phase Forecast (Baseline Projections)

These projections assume execution at the planned efficiency levels. They will be revised monthly from Phase 1 onwards when contract values and feasibility study costs are confirmed.

6.1 Phase 0 — Pre-Foundation

Item	Value
Duration	M1–M3 (Apr–Jun 2026)
BAC	¥0.25M
Planned end date	2026-06-30
Key gate	Founding team verbal commitments; village leadership informed

Item	Value
Current forecast end	On track for Jun 2026
Current forecast cost	¥0.16–0.22M

6.2 Phase 1 — Foundation

Item	Value
Duration	M4–M9 (Jul–Dec 2026)
BAC	¥5.5M
Planned end date	2026-12-31
Key gate	¥3–8M secured; legal entity registered; feasibility studies complete
Critical path	Forestry + Energy feasibility studies (procurement Q3 2026)
Primary risk	Government grant approvals often take 2–3 months longer than planned
Contingency plan	Reduce feasibility scope slightly; delay P2 start by 1–2 months

6.3 Phase 2 — Pilot Design

Item	Value
Duration	M10–M18 (Jan–Sep 2027)
BAC	¥22.25M
Planned end date	2027-09-30
Key gate	¥30–50M secured; detailed engineering complete; key permits in hand
Critical path	METI permitting + FIT registration (can take 6+ months)
Primary risk	Permit delays push Phase 3 start past Oct 2027
Contingency plan	Stage Phase 3 procurement independently of permit approvals for non-permit items

6.4 Phase 3 — Pilot Build

Item	Value
Duration	M19–M30 (Oct 2027–Sep 2028)
BAC	¥177M
Planned end date	2028-09-30
Key gate	Revenue operational; Phase 4 plan confirmed
Critical path	Building renovation → solar/battery install → EV charging → commissioning
Primary risk	Building structural findings requiring scope increase (currently low probability, high impact)

Item	Value
Contingency plan	Phase 3 WBS 5.9 includes ¥27M contingency (18%); Management Reserve of ¥25M backstop

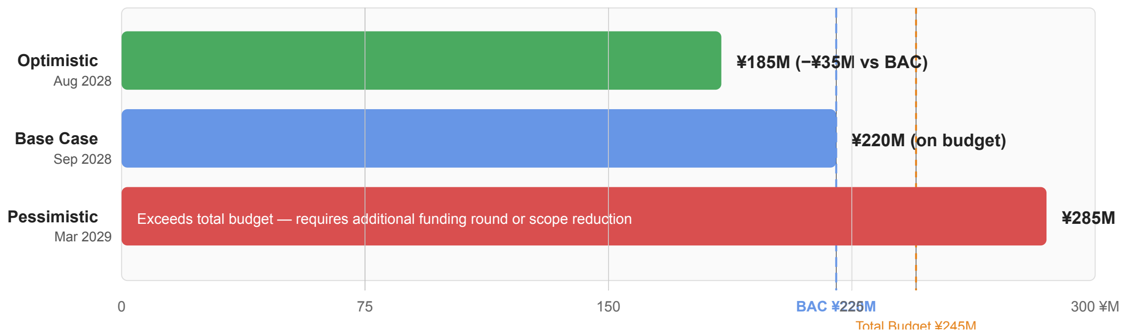
7. Three-Scenario Forecast

For a project with this level of regulatory and geographic complexity, planning in three scenarios is more honest than a single-point estimate.

Scenario	Description	EAC	End Date
Optimistic	All grants awarded on first application; building in good condition; no permit delays; lower-bound vendor quotes	¥185M	Aug 2028
Base Case	One major permit delay (+2 months); one grant deferred to Year 2; CPI ≈ 1.0; mid-range vendor quotes	**¥220M**	Sep 2028
Pessimistic	Two permit delays; building remediation required; government grant shortfall — draws on MR and beyond	¥285M	Mar 2029

> The scenario band is intentionally wider than the Rev 0 baseline. Phase 1 feasibility studies (M9) will narrow the range substantially once school structural condition, vendor quotes, and grant awards are confirmed.

Figure 4 — Three-Scenario EAC Comparison



> The pessimistic scenario at ¥285M **exceeds the total approved budget of ¥245M by ¥40M** and would trigger a Gate 3 funding decision: either (a) raise an additional funding round, (b) reduce Phase 3 scope (e.g., defer EV chargers or fiber upgrade), or (c) extend the schedule and stage construction. Absolute failure scenarios (community resistance, no funding at Gate 1) are still managed at the funding gate level, not the EVM level.

8. Performance Thresholds and Escalation

EVM metrics trigger review actions at the following levels. Thresholds are intentionally tighter in early phases when course-correction is cheapest.

Threshold	SPI or CPI	Action
Green	0.95 – 1.10	Normal reporting; no action required
Yellow	0.85 – 0.94	Representative Director notified; recovery plan drafted within 2 weeks
Orange	0.75 – 0.84	Board briefing required; recovery plan presented with options
Red	< 0.75	Formal board review; consider phase rescoping or Management Reserve draw

Schedule threshold override: If SPI < 0.90 at any funding gate checkpoint, the gate decision is automatically deferred pending a recovery assessment, regardless of CPI.

8.1 Liquidity control — cash balance is a hard rule

SPI/CPI measure *efficiency*; they do not tell you whether cash is in the bank. Because inflow arrives in lumps while spend is continuous (see `mitsue_cashflow_model.md`), the project separately tracks the **monthly cash balance = cumulative cash-in − cumulative cash-out** against a mandatory floor:

Liquidity floor	Rule
Phases 0–2	Balance must stay **≥ ¥5M** (≥ ¥3M absolute minimum at the M7 pinch, when founder capital is the only inflow)
Phase 3	Balance must stay **≥ one month of planned burn** (≥ ¥25M at the M25–M26 ¥25M/mo peak)
Breach forecast	If the 3-month rolling balance forecast projects a floor breach, the Representative Director triggers a tranche pull-forward or bridge draw **before** the breach month

These floors are **hard rules, not targets**: a projected breach is a stop/act condition equal in weight to a Red SPI/CPI. The monthly report tracks **balance, not just spend** (§9).

9. EVM Reporting Cadence

Frequency	Report	Recipients	Content
Monthly	EVM Status Update	Board, key funders	PV/EV/AC table; SPI/CPI; SV/CV; EAC update; **cash balance vs §8.1 liquidity floor + 3-month balance forecast**; issues
At each Gate	Gate Review Package	Board + major funders	Full EVM review; scenario forecast; go/no-go recommendation

Frequency	Report	Recipients	Content
Quarterly	Stakeholder Summary	Village government, advisors	High-level cost and schedule status; milestone progress
Annual	Year-End Review	All stakeholders	Full financial report; EVM audit; next-year baseline

Monthly reports are generated from OpenProject time tracking + invoicing. The first formal monthly EVM report is due **2026-06-30** (end of Phase 0). From Phase 1 onward, every monthly report carries the cash-balance line (cum-in − cum-out) against the §8.1 floor — the project tracks **balance, not just spend**.

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10. Assumptions and Constraints

Assumptions

- Project start date: 2026-04-01 (Month 1)
- All phase durations are calendar months, including Japanese holidays
- Feasibility studies are awarded to single vendors per study; no multi-vendor splits
- Phase 3 construction prices are based on 2026 Nara Prefecture rural construction indices, with a 15–25% rural mobilisation premium applied to commercial benchmarks for solar PV, EV charging, and seismic retrofit work
- Exchange rate assumptions (for Dutch/international corporate partnerships): ¥150/EUR
- Forestry operations are weather-dependent; M19–M23 baseline avoids typhoon season
- **The ¥28M funding gap is closed and disbursable before Gate 3 (M21).** Even in the best case where the ¥192M target stack lands as planned (none of it secured today), the cash balance runs negative at **M28 (Jul 2028)**, the shortfall equalling the gap exactly (see `mitsue_cashflow_model.md` §5)

Constraints

- BAC of ¥220M is a **ceiling**; Phase 3 cannot proceed without confirmed funding at Gate 3
- Management Reserve requires board approval for any drawdown
- **The §8.1 liquidity floor is binding:** the monthly cash balance may not fall below the floor; funding tranches must be scheduled to land before the spend they cover
- **Phase-3 funding tranches must be contracted 2–3 months ahead of their planned disbursement month** to absorb grant pay-in-arrears lag
- **A ~¥25M bridge facility** (bank line, Management Reserve draw, or signed-grant advance) must be arranged before Gate 3 as the working-capital shock absorber for the M22–M28 burn peak
- EVM data must be reported in JPY; foreign currency transactions converted at transaction-date rate
- The Phase 1 feasibility studies are the **single largest driver of overall project accuracy** — all EAC figures before M9 carry high uncertainty (±40%)

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11. Limitations and Honest Caveats

This is an **early-stage EVM plan** written at Month 2 of a 30-month project. The following limitations apply:

1. **CPI and SPI at Month 2 are not statistically reliable.** With only ¥0.08M of actual cost data, the performance indices reflect Phase 0 characteristics (personal time, low hard costs) rather than the project overall. Trust these numbers only after ¥2–3M of Phase 1 costs are confirmed.

2. **The Phase 3 budget range is wide.** The ¥120–290M range from the implementation plan reflects genuine uncertainty about building condition (¥30–100M school renovation range alone), equipment procurement

conditions, and grid connection costs. The EVM baseline uses ¥177M for WBS 5.0; the ¥27M Phase 3 contingency (18%) and ¥25M Management Reserve exist precisely to absorb this range.

3. **Funding gate gating is the primary control mechanism.** EVM monitors cost and schedule efficiency within a phase; the funding gates control whether the next phase begins at all. These two controls are complementary, not redundant.
4. **Phase 1 feasibility studies will substantially revise this baseline.** After M9, expect a formal baseline revision (Rev 2). The revised Phase 2 and Phase 3 budgets will be anchored to real survey data — particularly the school structural assessment and vendor quotes — not Japan-market benchmarks.

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12. Baseline Revision Policy

The PMB may be formally revised:

- After completion of Phase 1 feasibility studies (mandatory revision at M9)
- At any funding gate when confirmed funding differs from planned by more than ±20%
- When board-approved scope changes are incorporated
- When Management Reserve is drawn down

Each baseline revision is documented with: old baseline, new baseline, reason, approving authority, and date. Revisions are numbered:

- **Rev 0** (v1.1, Apr 2026) — Original BAC ¥168M, drafted from planning estimates
- **Rev 1** (v2.0, May 2026) — Current document. BAC ¥220M, reality-checked against real-world Japan benchmarks (solar, EV chargers, school seismic retrofit, forestry). Reason: pre-emptive correction in advance of Phase 1 feasibility studies.
- **Rev 2** (planned M9, Dec 2026) — Post-Phase 1 revision anchored to feasibility study results and vendor quotes. **Should also fold in any confirmed 地域脱炭素移行・再エネ推進交付金 amount** (2/3–3/4 subsidy on solar/battery/EV/private-wire capex, via village 官民連携) — the primary named path to closing the ¥28M–¥53M gap during Phases 2–3, which **must be secured and disbursable before Gate 3 (M21)**, not deferred to end-of-project (see `mitsue_cashflow_model.md` §5). Rev 2 must also **re-time the cashflow model to the live Gantt** in lockstep with the cost re-sync, and fold in confirmed 交付金 disbursement dates and any bridge-loan terms. Sources: https://policies.env.go.jp/policy/roadmap/grants/ · https://www.env.go.jp/content/900470616.pdf

Two additional FY2026 subsidy tracks (identified 2026-07-06) — both fund the biomass-power-plus-data-center model directly and should be evaluated at Rev 2:

- **経産省 GX地域共創補助金 (脱炭素電源地域貢献型投資促進事業)** — ¥210B FY2026 pool, integrated support for *decarbonized power + DC/factory* build-out; Round 1 Jul–Sep 2026, Round 2 autumn–winter 2026, 38 regions pre-screened. Strongest structural match (biomass power + compute co-located). Source: https://sustainablejapan.jp/2026/05/30/meti-gx-local/126036
- **環境省 データセンターのゼロエミッション化・地域共生加速化事業** — up to ¥1B/project on DC-linked renewable + battery equipment. ⚠ FY2026 application deadline **2026-07-03 (passed)**; target the FY2027 cycle. Source: https://sustainablejapan.jp/2026/06/07/japan-datacenter-renewable-energy/126312

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13. Founder Compensation & Sustainability

Early-phase founder time has been **largely unpaid**, representing the project's single most significant hidden dependency and top operational risk (see also the Implementation Plan's Risk Management table).

Target compensation. The Representative Director's target is **¥300,000/month (¥3.6M/yr)** — a defensible full-time-equivalent figure for rural Japan. This is a *target and norm*, **not a legal requirement**: a 一般社団法人 director may legally be unpaid; if paid, the amount must be set by **社員総会 (members' meeting) resolution**, not self-determined. For an NPO法人, pay for actual work performed by a director as staff (役員給与) is uncapped and separate from capped 役員報酬 for the role itself; once revenue is earned, salary must comply with 定期同額給与 tax rules.

> **Sources:** https://www.koueki-houjin.net/shadan/hosyu.html · https://www.koueki-houjin.net/shadan/kyuyo.html · https://ashiyakaikei.com/directors-npo/ · https://cliser.co.jp/non-profit/npo/159/

Phasing plan:

Period	Compensation	Funding source
Phase 0–1 (M1–M9)	Volunteer / nominal stipend	Founder capital; personal commitment
Phase 2 (M10–M18)	Partial stipend, ramping toward target	Layer-1 founding capital + first grants
Phase 3+ (M19+)	Full ¥300k/mo target	Phase 2 grants (¥30–50M) once confirmed

WBS 1.1a (¥5.0M over 30 months, ~¥167k/month average) covers the ramp inside the baseline. Reaching the full ¥300k/month run-rate from M10 is contingent on Phase 2 grant funding and will be confirmed at **Baseline Rev 2 (M9, December 2026)**.

Layer-1 founding capital is earmarked to cover the Phase 0–1 founder stipend so that early runway does not depend on entirely unpaid time.

The same phasing logic extends to the JP co-founder if that role is compensated — terms to be agreed and documented in the Founder Agreement before Phase 1.

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14. Phase 4 Forward Capital — Biomass Fuel-Prep & CHP (out of PMB)

This section is **informational and explicitly outside the ¥220M Performance Measurement Baseline**. It captures the scale-up capital for the biomass energy loop so funders can see the full trajectory; it does **not** alter the BAC. Phase 4 is funded by operating revenue + grants (see `mitsue_revenue_model.md`), not the PMB.

Sizing per the Forest Twin doubled-workforce baseline (~50% of forest under management, ~13,000 dry-t/yr fuel, 2×0.6 MWe). See `mitsue_forest_workforce_energy_plan.md` §5–6.

Bucket	Item	Forward capital (¥M)	Funding
F1	Biomass CHP gensets (2×0.6 MWe gasifier-genset)	~800	Revenue + FIT + 再エネ交付金
F2	Fuel-prep — active dryer, chip store, screening, handling, yard	~75–215 (mid ~120)	Revenue + grants; existing 牛峠工場 offsets ¥30–70
F3	Thermal store (molten-salt / packed-bed) + island-mode battery	*TBD (feasibility)*	国土強靭化 / 緊急防災債

Bucket	Item	Forward capital (¥M)	Funding
F4	Network connectivity to 神末797 (anchor compute) — NTT/leased-line or microwave-backup assessment + build	*TBD (feasibility)*	Revenue + grants
	Indicative Phase 4 forward capital	**~¥0.9–1.0 B + F4**	out-of-PMB

Relationship to the PMB: the Phase 3 pilot already carries a small forestry line (WBS 5.6, ¥25M, 5–10 ha harvest + replant). A **pilot-scale drying line** legitimately sits within that WBS; the full fuel-prep chain and the CHP fleet above are Phase 4. When Phase 4 is baselined (Year 2, after Phase 1 feasibility confirms accessible forest area and vendor quotes), these buckets become a separate Phase 4 PMB with its own BAC.

F4 is a separate connectivity problem from WBS 3.5/5.7. Those PMB line items (¥0.4M assessment + ¥10M upgrade) are scoped to the **old-school showcase DC** only. 神末797 (the anchor-compute site, co-located with the CHP) is a different, more remote location — its connectivity has not been assessed and should not be assumed to piggyback on the school's fiber build. Cost it alongside the grid 事前相談 already underway for 神末 (see `mitsue_fit_grid_check.md` §"Grid interconnection"), once NTT/ISP reach and a microwave-backup option are scoped.

> **Do not sum F1–F3 into the ¥245M total budget.** They belong to a later, separately funded phase and are shown here only for trajectory transparency.

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